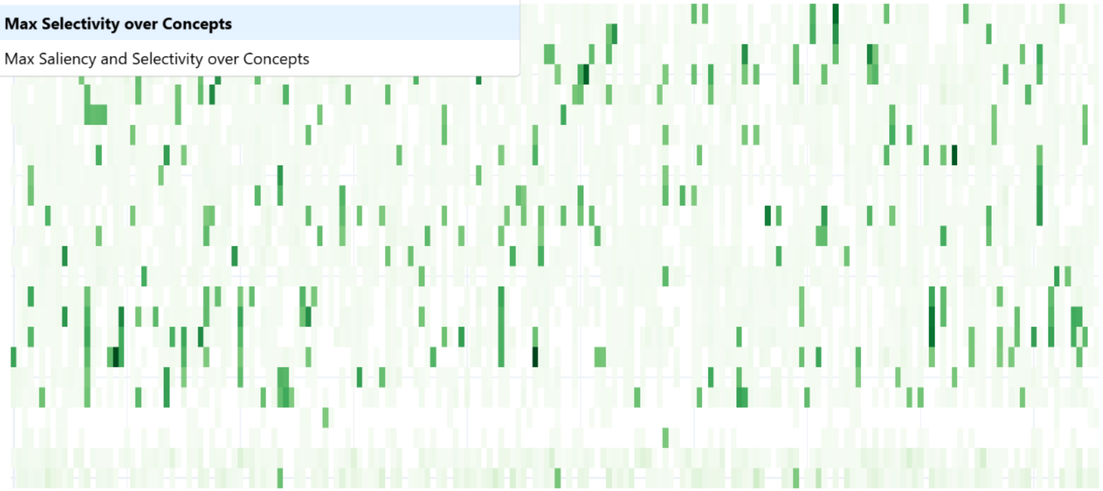
ConceptTracer

Metric: Max Selectivity over Concepts

- Max Saliency over Concepts
- **Max Selectivity over Concepts**
- Max Saliency and Selectivity over Concepts

All layers

Layers: 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0



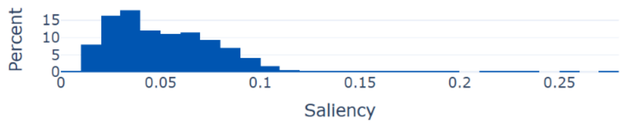
Neurons

Check one or more boxes to select layers (empty = ALL layers). Or click a heatmap cell to select a neuron. Note that insignificant values are excluded for improved speed and readability.

Selection → layers: ALL | neurons: ALL

All Concepts | Search concept

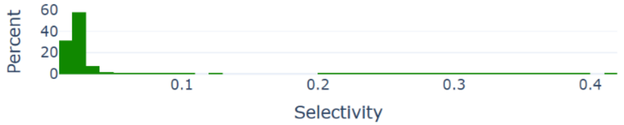
Saliency over Concepts



Top-k saliencies: 1 2 3 4 5 6 7 8 9 10

1. Concept E87, layer 0, neuron 162: saliency = 0.28, selectivity = 0.07
2. Concept E87, layer 0, neuron 109: saliency = 0.26, selectivity = 0.07
3. Concept E87, layer 0, neuron 147: saliency = 0.24, selectivity = 0.08
4. Concept E87, layer 0, neuron 73: saliency = 0.22, selectivity = 0.06
5. Concept E87, layer 0, neuron 128: saliency = 0.21, selectivity = 0.06

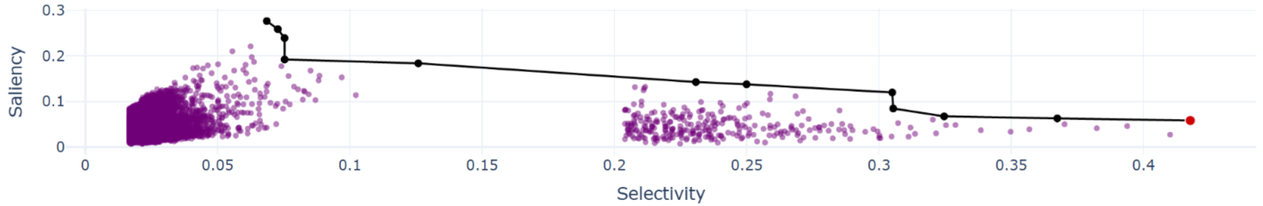
Selectivity over Concepts



Top-k selectivities: 1 2 3 4 5 6 7 8 9 10

1. Concept C, layer 6, neuron 92: saliency = 0.06, selectivity = 0.42
2. Concept C, layer 6, neuron 18: saliency = 0.03, selectivity = 0.41
3. Concept C, layer 16, neuron 166: saliency = 0.05, selectivity = 0.39
4. Concept I, layer 20, neuron 101: saliency = 0.04, selectivity = 0.38
5. Concept I, layer 23, neuron 145: saliency = 0.05, selectivity = 0.37

Saliency vs Selectivity



Top-k Pareto points: 1 2 3 4 5 6 7 8 9 10

1. Concept C, layer 6, neuron 92: saliency = 0.06, selectivity = 0.42
2. Concept I, layer 22, neuron 145: saliency = 0.12, selectivity = 0.31
3. Concept E87, layer 0, neuron 162: saliency = 0.28, selectivity = 0.07
4. Concept I, layer 21, neuron 70: saliency = 0.06, selectivity = 0.37
5. Concept E87, layer 0, neuron 109: saliency = 0.26, selectivity = 0.07